



Automated "Follow-Me Cart"

Using AI and Computer Vision for Disabled and Elderly Assistance



Abstract

The Automated Follow-Me Cart is an innovative assistive mobility solution designed to empower elderly and disabled individuals. Using AI and computer vision, the cart autonomously follows users, recognizes hand gestures, and avoids obstacles in real-time. It is durable, portable, and adaptable to various environments such as homes and stores.

The integrated mobile app enhances usability by enabling battery monitoring, cart content tracking, store mapping, and remote control. This project combines advanced technology and engineering to promote independence, convenience, and inclusivity.

Who We Are

We are a team of ambitious engineers from the Faculty of Engineering, Mansoura University, specializing in Mechatronics. Our mission is to leverage modern technology to create solutions that make life easier for those in need.

The Challenge

Mobility issues among the elderly and disabled significantly limit independence and quality of life. Traditional aids often lack adaptability and ease of use.

The Solution

The "Follow-Me Cart" combines artificial intelligence, computer vision, and durable mechanical design to offer hands-free mobility assistance.

Features

Key Features of the Cart



AI-Powered Gesture Recognition
Hands-free, intuitive control.



Obstacle Detection & Avoidance
Real-time safety for smooth navigation.



Autonomous User Following
Enhances mobility and independence.



Durable & Portable
Designed for various terrains with extended usability.

Core Components



Motors
Provide smooth movement with a capacity of up to 62.1 kg



Camera & AI Unit
Enables hand gesture tracking using advanced algorithms.



Sensors
Infrared and ultrasonic sensors ensure obstacle-free operation

Technical Overview

How It Works



Mechanical Design
CAD-based design for durability and ease of assembly



AI Integration
Processes real-time data to follow users and avoid obstacles.



Power System
Rechargeable battery system ensures seamless operation

Innovative Technologies

1. NVIDIA Jetson platform for AI processing.
2. Arduino Nano for auxiliary functions.
3. Custom algorithms for gesture recognition and path planning.

Abstract

The Follow-Me Cart Mobile App enhances the cart's functionality by offering real-time battery monitoring, store mapping, item tracking with cost calculations, and remote control. Designed for ease of use, it provides a smarter, more convenient, and hands-free experience for elderly and disabled users.

Key Features of the App

-  **Battery Monitoring**
Track battery status anytime
-  **Store Mapping**
Navigate with real-time mapping
-  **Item Tracking**
Navigate with real-time mapping
-  **Remote Control**
Move the cart effortlessly via the app.



How It Works

The mobile app complements the cart, offering full visibility and control, ensuring convenience, safety, and efficiency.

- Connect**
Sync with the cart via Bluetooth
- Monitor**
View battery, cart space, and navigation details.
- Control**
Remotely manage cart movement.

